Semi-Analytical Modelling of Thin-Walled Structures: Theory and Practice

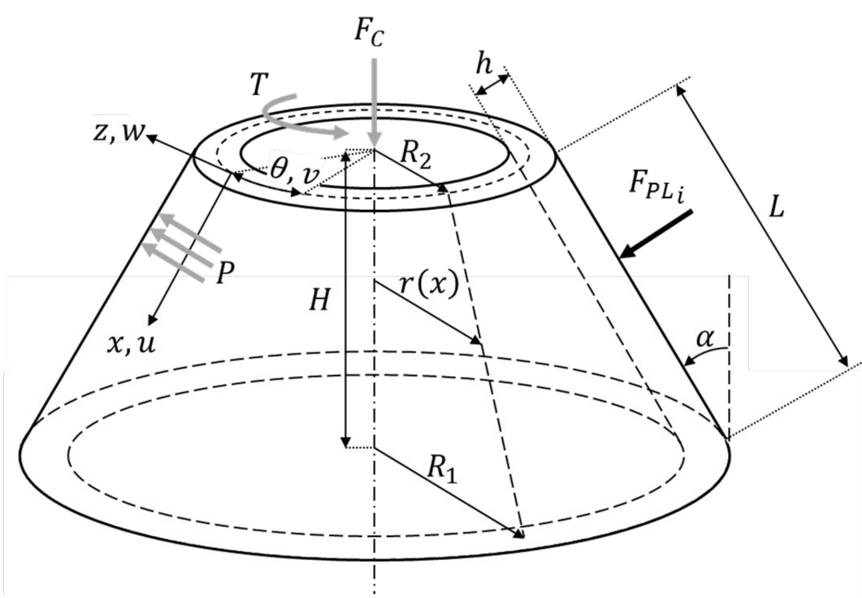
2026 Edition

$$\delta\Pi = \delta U + \delta V = \{\delta c\}^T\{R\}$$
$$\delta^2\Pi = \delta(\delta U + \delta V) = 0$$

$$u(x,\theta,z,t) = u_0(x,\theta,t) + z\phi_x(x,\theta,t)$$
$$v(x,\theta,z,t) = v_0(x,\theta,t) + z\phi_\theta(x,\theta,t)$$
$$w(x,\theta,z,t) = w_0(x,\theta,t)$$
$$\phi_x = -w_{,x}$$
$$\phi_\theta = -\frac{1}{r}w_{,\theta} + \delta_1\frac{v}{r}\cos\alpha$$

$$\delta U = \int_{\mathcal{V}} \{\delta\varepsilon\}^{\mathrm{T}}\{\sigma\}\,\mathrm{d}\mathcal{V}$$
$$\delta V = -\{\delta c\}^T\left(\{F_{ext_0}\} + \lambda\{F_{ext_\lambda}\}\right)$$

$$\{u\} \approx [g]\{c\}$$



Saullo G. P. Castro